

- Q.27 Write a note on ultrasonic therapy unit.
- Q.28 Explain the need and importance of CSSD in hospitals.
- Q.29 Explain sterilization and its classification.
- Q.30 Explain Incubator with block diagram.
- Q.31 Write a short note on pain relief through electric stimulation.
- Q.32 Explain short wave diathermy with diagram.
- Q.33 Explain therapeutic instrumentation.
- Q.34 Explain ultrasonic washer with block diagram.
- Q.35 Explain safety aspects of electro-surgical machine.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain LASER with their characteristics and classification with block diagram.
- Q.37 Explain haemodialysis machine with classifications with block diagram.
- Q.38 Explain TENS machine with diagram.

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Roll No.

5th Sem / Med. Eltx

Subject:- Therapeutic Medical Instrumentation (TMI)

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A major safety hazard is diathermy is
- Noise
 - Burns due to current leakage
 - low voltage
 - cooling failure
- Q.2 Which of the following used for newborn temperature control
- Diathermy
 - Incubator
 - ECG
 - Ultrasonic washer
- Q.3 Laser are used in
- Neurology
 - Cardiac imaging
 - Surgical treatments
 - BP measurements
- Q.4 Physiotherapy using electrical stimulation called
- Diathermy
 - Electro-therapy
 - Hydrotherapy
 - Ultrasound

- Q.5 Short wave diathermy uses
- Low frequency current
 - High frequency EM Waves
 - Microwave
 - IR ways
- Q.6 TENS used for
- Temperature regulation
 - Nerve stimulation
 - Blood pressure
 - All of the above
- Q.7 Which type of current used in ELISA Reader
- AC
 - DC
 - Optica; Signals
 - Thermal pulses
- Q.8 Which type of dialyser has multiple hollow fibers
- Coil
 - Parallel
 - Hollow fiber
 - Spiral
- Q.9 Which component check conductivity of dialysis fluid
- Blood pump
 - Sensor
 - flow meter
 - Conductivity monitor
- Q.10 Which electrode used for coagulation in diathermy
- Bipolar
 - Monopolar
 - Return electrode
 - negative electrode

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name one method of sterilization.
- Q.12 Expand CSSD
- Q.13 Which department handle sterilized handles in hospital.
- Q.14 Ideal temperature for dry heat sterilization.
- Q.15 Expand TENS
- Q.16 Name one chemical sterilizing agent
- Q.17 Which indicator used to check sterilization
- Q.18 Name of current used in TENS
- Q.19 Which organ support haemodialysis?
- Q.20 Main function of artificial kidney?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Enlist different equipment used in Laser therapy.
- Q.22 Explain basic principle and operation of LASER.
- Q.23 Name different types of currents used in electro-therapy.
- Q.24 Write a note on micro-wave diathermy.
- Q.25 Explain Laser characteristics & their medical application.
- Q.26 Explain ELISA Reader.